



OWNER SETUP & SELLING GUIDE

Duka POS

Everything you need to launch your POS, charge shops automatically every month, and sign up the clients already waiting — written for a non-technical owner.

THE SHORT VERSION

Put it online (1 hr) → connect M-PESA → set your prices → onboard each shop on a 14-day free trial → the system charges them monthly and locks the POS if they don't pay.

Duka POS — offline-first Point of Sale for Kenyan shops, sold as a monthly subscription. Built for Kenya. Ready for Africa.

What's in this guide

1. **Part 1 — Understand your business** **read first** — what you're selling and how money reaches you.
2. **Part 2 — Put the system online** **1 hour** — the app and the biller, step by step.
3. **Part 3 — Connect M-PESA & messages** **a few days to approve** — get your keys.
4. **Part 4 — Set your prices** — the plans and how to change them.
5. **Part 5 — How shops are charged automatically** — the four methods, and the "lock".
6. **Part 6 — Sign up a new shop (onboarding)** — the exact steps per client.
7. **Part 7 — Selling it** — a simple pitch and answers to common questions.
8. **Part 8 — Running it day to day** — what to check, what to do when a shop won't pay.
9. **Part 9 — Staff logins & your control room** — roles per shop, and your Super-Admin portal.
10. **Go-live checklist & Quick reference** — the one-page cheat sheets.

You do NOT need to understand code

Everywhere below, you only fill in web forms and copy-paste a few values. When a box says `LIKE_THIS`, that's just a label you'll paste a value next to — nothing to "program".

PART 1

Understand your business

You are the **platform owner**. Shops (dukas) use your POS to sell, and pay you a monthly fee to use it.

There are two pieces, and both are already built for you:

- **The POS app** — what each shop uses to sell, track who owes them (debt), and send WhatsApp/SMS reminders. Works on any phone, tablet or computer, even offline.
- **The biller (helper service)** — your money machine. It charges each shop every month by M-PESA, and if a shop doesn't pay, it automatically **locks their POS** until they do.

How you make money: every shop pays you monthly (from KES 1,900 to KES 24,900 depending on their size). The money lands in **your** M-PESA Till/Paybill. You never chase anyone — the system charges them and enforces payment by holding the POS.

Put the system online

Two free sign-ups: one for the app (Vercel), one for the biller (Render). Your code is already on GitHub — you're just switching it on.

2A • The POS app (Vercel) — ~10 min

1

Create a free account

Go to **vercel.com** → **Sign Up** → **Continue with GitHub**. Approve when GitHub asks.

2

Import the project

Add New... → **Project** → find `duka-pos` → **Import** → **Deploy**. Wait ~2 minutes for "Congratulations".

3

Get your link

You'll get an address like `https://duka-pos-xxxx.vercel.app`. This is the POS. Bookmark it.

2B • The biller (Render) — ~10 min

1

Create a free account

Go to **render.com** → **Sign up** → **Continue with GitHub**.

2

Create from blueprint

New + → **Blueprint** → choose `duka-pos` → **Apply**. You'll get an address like `https://duka-pos-backend.onrender.com`. Copy it.

3

Connect the app to the biller

In **Vercel** → **your project** → **Settings** → **Environment Variables**, add: name `VITE_API_URL`, value = your Render address. Then **Deployments** → **Redeploy**.

2C • Give shops a memory that can't be wiped (database)

By default the biller remembers shops in a simple file. For real business, use a free database so nothing is ever lost:

1

Create a free database

Go to **supabase.com** → new project. Open **Project Settings** → **Database** → **Connection string (URI)** and copy it.

2**Give it to the biller**

In **Render** → **your service** → **Environment**, add `DATABASE_URL` = that connection string. Render restarts and creates the table by itself.

How to know it worked

Open `https://YOUR-backend.onrender.com/api/health` in a browser. It should say `"storage": "postgres"` once the database is connected (or `"memory"` before that).

Connect M-PESA & messages

These let you take real money and send real WhatsApp/SMS. Start the M-PESA application early — approval is the slowest part. Until keys are added, everything runs in a safe demo mode.

3A • M-PESA (to collect subscription money)

- You need a **Lipa na M-PESA Till or Paybill** (from Safaricom / M-PESA Business) — this is where shops' payments land.
- Then register on the **Safaricom Daraja** developer portal (`developer.safaricom.co.ke`) and create an app to get your keys.

Paste these into Render → Environment

`MPESA_CONSUMER_KEY` `MPESA_CONSUMER_SECRET` `MPESA_SHORTCODE` (your Till/Paybill)
`MPESA_PASSKEY` `MPESA_ENV=production`

3B • Messages (so reminders send by themselves)

TO TURN ON	SIGN UP AT	PASTE INTO RENDER
Automatic SMS	<code>africastalking.com</code>	<code>AT_API_KEY</code> , <code>AT_USERNAME</code>
Automatic WhatsApp	<code>business.facebook.com</code>	<code>WHATSAPP_TOKEN</code> , <code>WHATSAPP_PHONE_ID</code>

Order of operations

You can go live and start selling **before** messages are approved — shops just tap "Remind" by hand (still one tap, still includes your pay details). Turn on automatic sending whenever the keys arrive.

Where each value goes

App settings (like `VITE_API_URL`) go in **Vercel**. Money/message keys (like `MPESA_*`) go in **Render**. That's the only rule to remember.

PART 4

Set your prices

These are the monthly fees shops pay you. They're benchmarked to the Kenyan POS market and can be changed.

PLAN	BEST FOR	MONTHLY	YEARLY (2 MONTHS FREE)
Starter (Duka Ndogo)	1 small shop, up to 2 staff, ~500 products	KES 1,900	KES 19,000
Standard (Duka)	1 busy shop, up to 6 staff, ~3,000 products	KES 4,900	KES 49,000
Growth (Duka Biashara)	up to 3 branches, 20 staff	KES 9,900	KES 99,000
Chain (Duka Mtandao)	many branches + warehouse	KES 24,900	KES 249,000

Every shop starts on a **14-day free trial** with full features — no money needed to begin.

Want to change a price?

Prices live in one small file. Ask your developer (or a follow-up session) to edit `src/lib/plans.ts` (the app) and `server/lib/plans.js` (the biller) — change the `price` numbers, save, and it redeploys. Keep the two files matching.

Pricing tip for your waiting clients

Offer the annual plan to anyone who can pay up front — you get 10 months of cash today and they save 2 months. For the rest, monthly on auto-charge keeps cash flowing without you chasing.

How shops are charged automatically

M-PESA can't silently take money like a card — so "automatic" works in one of four ways. All are built in; you choose which to offer.

METHOD	HOW AUTOMATIC	WHAT THE SHOP DOES
1. Auto STK push on the due date	Semi — you charge, they approve	Enters M-PESA PIN when the prompt pops up
2. Paybill auto-detect	Automatic detection	Pays your Paybill like any bill; system renews them
3. M-PESA Ratiba (standing order)	Fully automatic	Approves once; Safaricom debits every month by itself
4. Card (optional)	Fully automatic	Enters card once

The part that makes shops actually pay: if a shop is late, the POS **holds** — day 1-7 it still works (with reminders), day 8-14 it can view but **can't sell**, day 15+ it's locked with a "Pay to reactivate" screen. The moment they pay, it unlocks. This is automatic and already built.

What you do

- Leave the biller's **scheduler on** (it is by default) — every due shop gets charged automatically each cycle.
- For the most hands-off shops, offer **Ratiba** at sign-up so it debits with zero taps.
- You can watch it happen safely in demo mode before going live (see the "Try it" box in `BILLING.md`).

Recommended setup

Ratiba for shops who accept it → STK push on the due date for everyone else → Paybill auto-detect as a safety net → the POS-hold to enforce it. That combination is automatic for most and self-enforcing for the rest.

Sign up a new shop

Do this with each client. It takes about 15 minutes and they can sell the same day.

1 Send them the link

Give them your Vercel POS link. On their phone they open it in Chrome/Safari and choose **Add to Home Screen** — now it's an app icon that works offline.

2 Set up their shop

In **Settings**: shop name, phone number (this is their billing identity), location, and their own **M-PESA Till/Paybill** (so their customers can pay them — separate from your platform Till).

3 Pick their plan

On the **Billing** screen, use "**Find your fit**" (enter their shops/staff/products) — it recommends Starter, Standard, Growth or Chain. Their **14-day free trial** starts automatically.

4 Add their products

In **Stock**, add items (name, price, stock). Then **Settings** → **Data** → **Clear everything** once to remove the demo items and start clean.

5 Show them the essentials

Selling on the **Sell** screen, recording credit (mkopo) under a customer, and tapping **Remind** on the **Debts** screen to WhatsApp/SMS a debtor.

6 They're live

On day 10 of the trial they'll be nudged to pay; on day 14 the trial ends and the auto-charge begins. Nothing more for you to do.

Their phone number matters

The number you enter in a shop's Settings is how the biller identifies and charges them. Use the owner's real M-PESA number.

Selling it to your clients

You already have shops waiting. Here's a simple way to pitch and close.

The 20-second pitch

"It's a POS that works even when your internet is down, and it automatically reminds customers who owe you — by WhatsApp and SMS — with your M-PESA number in the message so they can pay on the spot. You get paid faster. It's KES 1,900 to KES 4,900 a month, and the first 14 days are free."

The three things shop owners care about

- **"Will it work when Safaricom is down?"** Yes — it keeps selling offline and syncs later.
- **"Can it help me collect debts (mkopo)?"** Yes — it tracks who owes and sends reminders with your pay details. This alone usually pays for the subscription.
- **"Is it hard?"** No — a sale takes seconds, and it's on their phone.

Handling "it's too expensive"

- Point at the debt feature: recovering even one or two forgotten debts a month covers the fee.
- Offer the **Starter** plan (KES 1,900) to very small shops.
- Offer **annual** — they pay for 10 months, use 12.
- Let the **14-day free trial** do the selling — they try it risk-free.

A smooth first week

Sign up 2–3 shops, sit with each for 15 minutes, and make sure they process one real sale and send one real reminder while you're there. Happy early users bring you the next ones.

PART 8

Running it day to day

- **Watch your shops:** open <https://YOUR-backend.onrender.com/api/admin/tenants> to see every shop, their plan, and whether they're paid, in grace, on hold, or suspended.
- **When a shop doesn't pay:** do nothing — the system moves them from grace → hold → suspended and keeps prompting them to pay. They reactivate instantly when they do.
- **When a shop pays:** nothing to do — their POS unlocks and an invoice is recorded automatically.
- **New client:** follow Part 6. **Price change:** see Part 4.

Two things to keep healthy

- The free Render biller "sleeps" when idle. For punctual monthly charging, upgrade Render to a small paid tier (a few dollars/month) so it's always awake.
- Keep your M-PESA and database keys safe. Never share them or post them anywhere.

Costs summary

PIECE	COST
The POS app (Vercel)	Free
The biller (Render)	Free to start; ~few \$/mo to stay always-on
Database (Supabase)	Free tier
SMS (Africa's Talking)	~KES 0.80 per SMS sent
WhatsApp (Meta)	Free / low per conversation
M-PESA (Daraja)	Free to connect; normal Till/Paybill rates

PART 9

Staff logins & your control room

Two kinds of access: staff **inside each shop**, and you **across all shops**.

9A · Staff on the shop device

Each shop's device is **PIN-locked**. Staff tap their name and enter a PIN to sign in; the **lock** button switches user. The owner adds staff in **Settings → Staff & roles**.

ROLE	CAN DO	CANNOT DO
Owner	Everything	—
Manager	Sell, stock, reports, discounts, debts	Billing, manage staff
Cashier	Sell & serve customers, view/chase debts	Edit stock/prices, reports, settings, discounts

Setting PINs

Give each person their own PIN and role. Demo PINs to start: Owner 1234 , Cashier 0000 — change these before real use (Settings → Staff & roles).

9B · Your Super-Admin portal (all shops)

You get a private control room to run the whole business:

1

Set your admin password

In **Render → your service → Environment**, add `ADMIN_TOKEN` = a long secret you make up.

2

Open your portal

Go to `https://YOUR-backend.onrender.com/admin` and paste that token.

3

Run the business

See every shop with its plan, status and renewal date, your **estimated monthly revenue (MRR)**, and buttons to **charge a shop now** or **run a billing sweep**.

Keep this token secret

Anyone with the admin token can see and manage all your shops. Don't share it. If it leaks, change `ADMIN_TOKEN` in Render.

If the app itself has a problem

You own the code and the servers, so any issue can be fixed and redeployed — by you with help, or in a follow-up session. Your shops' data is safe in your database (Part 2C).

Go-live checklist

Set up (once)

- ☐ Vercel account + `duka-pos` deployed
- ☐ Render account + biller deployed (Blueprint)
- ☐ `VITE_API_URL` set in Vercel → redeployed
- ☐ Supabase database + `DATABASE_URL` in Render
- ☐ M-PESA Till/Paybill + Daraja keys in Render
- ☐ (Optional) SMS / WhatsApp keys in Render
- ☐ `ADMIN_TOKEN` set in Render → `/admin` portal opens
- ☐ Changed the demo staff PINs (1234 / 0000)
- ☐ `/api/health` shows storage + providers OK

Per shop

- ☐ Sent them the app link
- ☐ Settings: name, phone, their M-PESA
- ☐ Plan chosen (trial started)
- ☐ Products added, demo data cleared
- ☐ Showed sale + reminder
- ☐ Confirmed they appear in `/api/admin/tenants`

Quick reference

I WANT TO...	WHERE
See all my shops & who's paid	<code>/api/admin/tenants</code> on your backend
Change app settings	Vercel → Settings → Environment Variables
Change money/message keys	Render → your service → Environment
Change prices	<code>src/lib/plans.ts</code> + <code>server/lib/plans.js</code>
Read the technical detail	<code>DEPLOYMENT.md</code> , <code>BILLING.md</code> , <code>README.md</code> in your repo

You're ready

Put it online, connect M-PESA, set your prices, and start onboarding the shops that are waiting. The system sells, tracks debts, charges monthly, and enforces payment — so you can focus on signing up more dukas. Karibu, and good luck. 🇰🇪